Chemistry

Concise Lecture Notes

Chemical Bonding

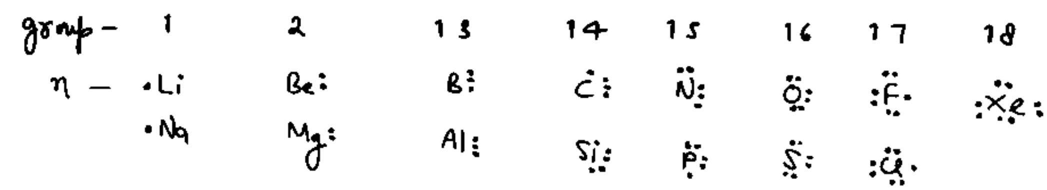
Key topics covered in the Class:

- Chemical Bonding (Ionic, Covalent, Metallic)
- VSEPR Theory and Hybridisation
- Molecular Orbital Theory (MOT)
- Dipole Moment and Intermolecular Forces

Valence Electron & Lewis Symbol

- The number of dots in a Lewis symbol equals the number of valence electrons.
- Bond formation is an exothermic process (releases energy, ΔH is negative).

No of dots = no of valence electrons



Bond

- **Strong Bonds:** High bond energy (~200 kJ/mole); includes Ionic, Covalent, Co-ordinate, and Metallic bonds.
- **Weak Bonds:** Low bond energy (4–20 kJ/mole); includes Hydrogen bonds and Van der Waals forces.

Octet Rule

- Atoms combine by transferring or sharing electrons to achieve a set of 8 electrons (octet) in their valence shell.
- Ionic bonding involves electron transfer; covalent bonding involves electron sharing.

Limitations of Octet rule

- **Incomplete Octet:** Hypo-valent molecules with less than 8 electrons (e.g., LiCl, BF_3).
- **Odd Electron Molecules:** Species with unpaired electrons (e.g., NO, NO_2).

- **Expanded Octet:** Hyper-valent molecules with more than 8 electrons due to available d-orbitals (e.g., PF_5, SF_6).

Covalency

- Covalency is the number of bonds formed by the sharing of electrons by an element.
- Group Valency equals the number of electrons in the valence shell.

Existence & Non – existence

- **Unavailability of d orbitals:** 2nd period elements (N, O, F) cannot expand their octet (e.g., NCl_5 does not exist).
- **Steric Crowding:** Molecules may not exist if the central atom is too small to accommodate large surrounding atoms (e.g., SI_6 does not exist).

d orbital contraction

- Highly electronegative substituents (like F) cause partial positive charge on the central atom, contracting d-orbitals to make them available for bonding.
- This allows molecules like PF_5 to exist, while PH_5 does not.

Formal Charge

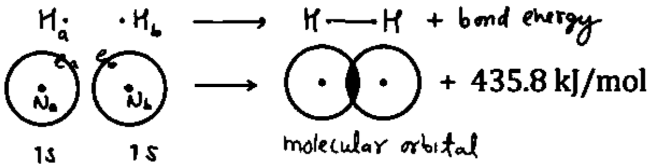
- Used to determine the distribution of charge within a molecule.
- Formula: Formal Charge = (Valence e^-) – (Non-bonding e^-) – (Bonding $e^-/2$).

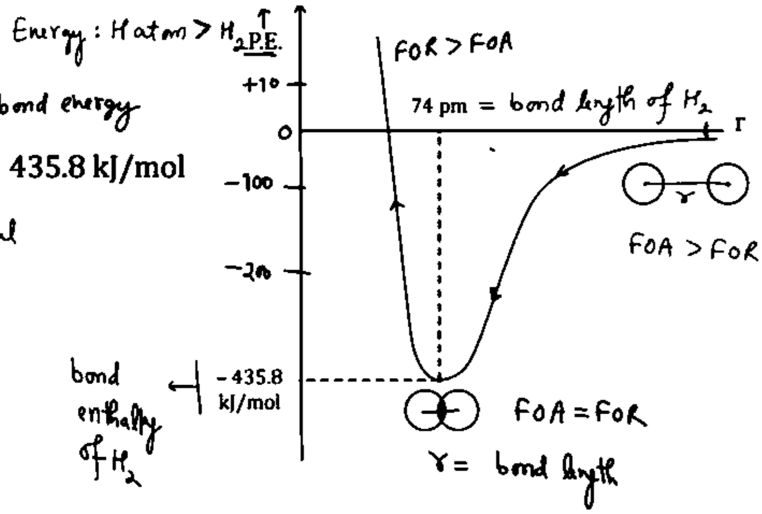
Resonance

- Involves the delocalization of pi (π) electrons, resulting in a Resonance Hybrid structure which is more stable than individual canonical forms.
- **Bond Order Formula:** Bond Order $= 1 + \frac{\text{Number of } \pi \text{ bonds}}{\text{Number of } \sigma \text{ bonds}}$

Valence Bond Theory

- Bonds form when attractive forces overcome repulsive forces as atoms approach each other.
- Potential Energy is minimal at the specific internuclear distance known as the bond length.





FoA : (i) $N_a - e_b$
(ii) $N_b - e_a$

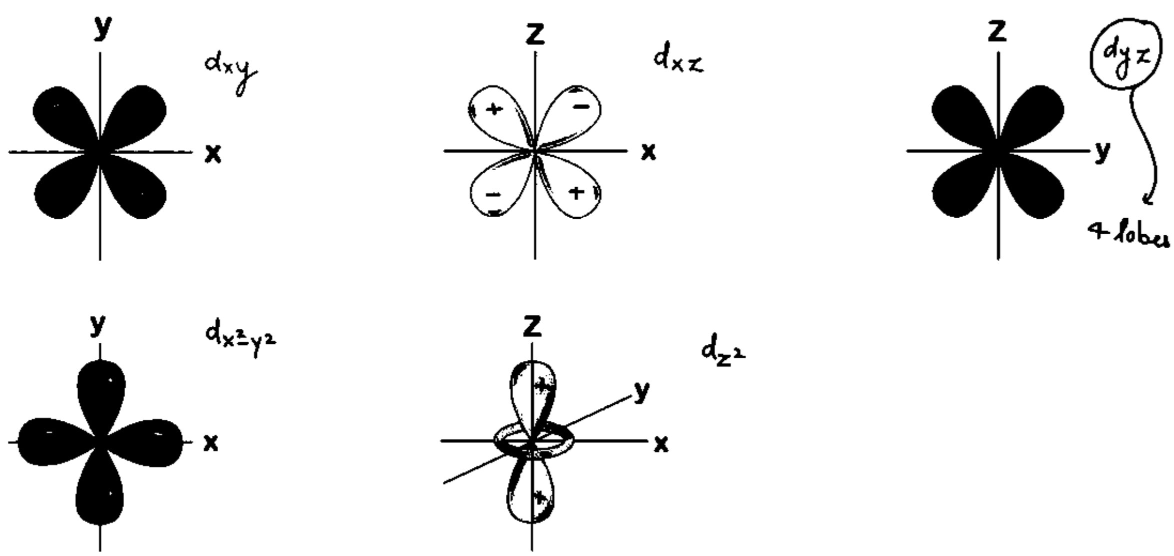
FoR : (i) $e_a - e_b$
(ii) $N_a - N_b$

Nature & Phase of Atomic Orbital

- **s-orbital:** Spherical, non-directional, 1 lobe.
- **p-orbital:** Dumbbell shape, directional, 2 lobes with positive/negative phases.

d-Orbital

- Contains 4 lobes (except d_{z^2} which has 2 lobes and a ring).
- Axial orbitals ($d_{x^2-y^2}$, d_{z^2}) lie on the axes; non-axial (d_{xy}, d_{yz}, d_{zx}) lie between axes.



Positive Overlap

- Occurs when orbitals of the same phase/sign overlap (+/+ or −/−), leading to bond formation.

Sigma Bond

- Formed by head-on (axial) overlapping along the molecular axis.
- Extent of overlapping (Bond Strength): $p-p > s-p > s-s$.

Pi Bond

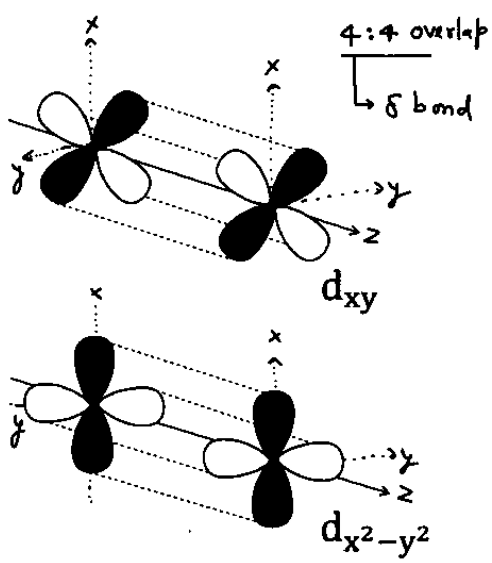
- Formed by sideways (lateral) overlapping perpendicular to the molecular axis (2:2 overlap).
- Weaker than sigma bonds; can be $p\pi - p\pi$ or $p\pi - d\pi$.

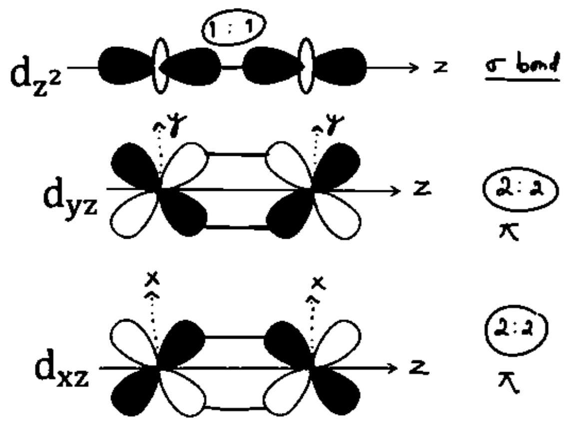
Strength of Overlapping

- When shell numbers are different, smaller shells overlap stronger ($1s-1s > 2s-2s$).
- When shell numbers are the same, directional orbitals overlap stronger ($2p-2p > 2s-2p > 2s-2s$).

Bonding in d orbitals

- d-orbitals can form sigma bonds (e.g., d_{z^2} axial overlap), pi bonds, or delta bonds (4:4 overlap).





	σ	π	δ
s	✓	×	×
p	✓	✓	×
d	✓	✓	✓

Valence Shell Electron Pair Repulsion Theory (VSEPR)

- Electron pairs around a central atom arrange themselves to minimize repulsion.
- **Repulsion Order:** Lone Pair-Lone Pair > Lone Pair-Bond Pair > Bond Pair-Bond Pair.

Order of repulsive interaction

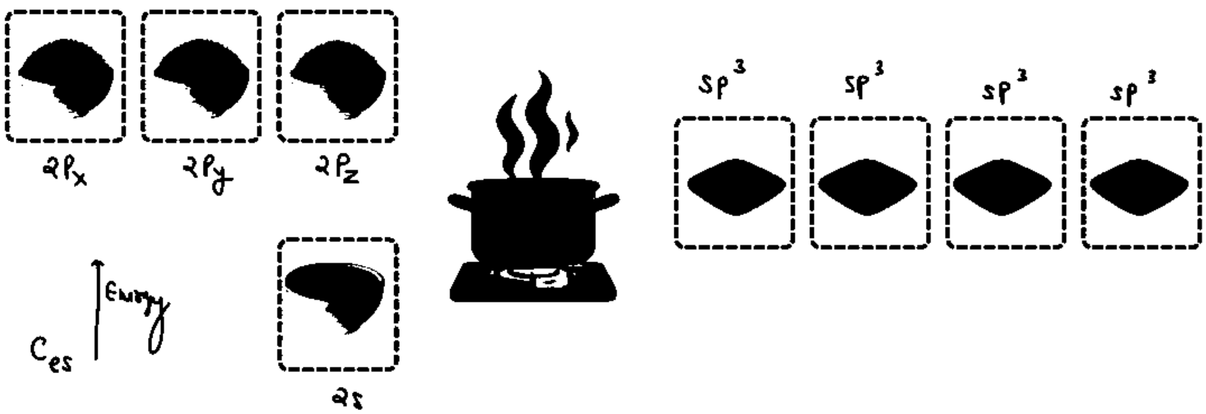
- **Steric Number (SN):** SN = (Number of Sigma Bonds) + (Lone Pairs).
- Used to determine geometry: SN=2 (Linear), SN=3 (Trigonal Planar), SN=4 (Tetrahedral).

Need of new concept (Hybridisation)

- VBT could not explain why all bonds in CH_4 are identical in length and energy.
- Experimental fact: All C-H bonds in Methane are equal with a bond angle of $109°28'$.

Hybridisation

- Inter-mixing of pure atomic orbitals of comparable energy to produce new, equivalent hybrid orbitals.
- The number of hybrid orbitals generated equals the number of atomic orbitals mixed.



Drago's Compounds

- Elements like P, As, Sb, S, Se, Te (3rd period and below) attached to less electronegative elements (H) do not undergo hybridisation.
- Pure p-orbitals are used for bonding, resulting in bond angles near $90°$ (e.g., PH_3).

Steric Number Rule

- SN = 5 (sp^3d): Trigonal bipyramidal geometry.
- SN = 6 (sp^3d^2): Octahedral geometry.
- SN = 7 (sp^3d^3): Pentagonal bipyramidal geometry.

Bond angle & % s character

- **Relationship:** Orbital electronegativity and bond angle increase with % s-character ($sp > sp^2 > sp^3$).
- **Orbital Length:** Increases as % s-character decreases ($sp < sp^2 < sp^3$).

Bond Length & % s character

- More % s-character leads to shorter bond lengths.
- Example: $C-H$ bond length is shorter in alkynes (sp) than in alkanes (sp^3).

PCl_5

- Hybridisation is sp^3d.
- Axial bonds are longer and weaker than equatorial bonds due to greater repulsion from equatorial pairs.

Bent's Rule

- Lone pairs prefer hybrid orbitals with greater % s-character.
- More electronegative atoms prefer hybrid orbitals with smaller % s-character (more p-character).

Hybridization in Odd Electron Species

- If the side atoms are highly electronegative, the odd electron counts in the Steric Number (e.g., NO_2, CF_3).
- If side atoms are not electronegative, the odd electron is not counted (e.g., CH_3).

No. of $p\pi - p\pi$ bonds / No of $p\pi - d\pi$ bonds

- Elements in the 2nd period (C, N, O) form only $p\pi - p\pi$ bonds.
- Elements in the 3rd period and below (P, S, Cl) form $p\pi - d\pi$ bonds.

Equivalent hybrid orbitals

- If the central atom has no lone pairs and is attached to identical atoms, the hybrid orbitals are equivalent (e.g., CH_4).
- If lone pairs exist or side atoms differ, orbitals are non-equivalent (e.g., H_2O).

Karishma of % s character in Bond angle

- As % s-character increases, the bond angle increases.
- Lone pairs consume s-character, reducing the s-character available for bonds, thus decreasing the bond angle $(CH_4 > NH_3 > H_2O)$.

Bond Angle

- **Rule 1**: Determine hybridisation $(sp > sp^2 > sp^3)$.
- **Rule 2**: If hybridisation is the same, more lone pairs = smaller bond angle.
- **Rule 3**: If lone pairs are equal, higher electronegativity of the central atom = larger bond angle.

Dipole Moment

- Measure of polarity in a bond; vector quantity ($\mu = q \times d$).
- **Non-polar**: Symmetrical molecules (e.g., CO_2, CCl_4) have $\mu_{resultant} = 0$.
- **Polar**: Unsymmetrical molecules or those with lone pairs (e.g., H_2O, NH_3) have $\mu \neq 0$.

Need of New Theory : MOT

- VBT wrongly predicts O_2 is diamagnetic (all paired electrons).
- Experiments show O_2 is paramagnetic (unpaired electrons), which MOT explains correctly.

Linear Combination of Atomic Orbitals.

- **Bonding Molecular Orbital (BMO)**: Formed by addition of wave functions ($\psi_A + \psi_B$); lower energy, stable.
- **Antibonding Molecular Orbital (ABMO)**: Formed by subtraction ($\psi_A - \psi_B$); higher energy, unstable, denoted with *.

Conditions for the Combination of Atomic Orbitals

- Combining orbitals must have comparable energies (e.g., 1s with 1s).
- Orbitals must have the same symmetry about the molecular axis.

$n \leq 14$

- Due to s-p mixing, the energy of σ_{2p} increases.
- **Energy Order**: $\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \pi_{2p} < \sigma_{2p} < \pi_{2p}^* < \sigma_{2p}^*$.

$n > 14$

- No s-p mixing occurs.
- **Energy Order**: $\sigma_{1s} < \sigma_{1s}^* < \sigma_{2s} < \sigma_{2s}^* < \sigma_{2p} < \pi_{2p} < \pi_{2p}^* < \sigma_{2p}^*$.

Ionic Bonding

- Formed by the transfer of electrons from a metal (cation) to a non-metal (anion).
- It is non-directional in nature.

Lattice Energy

- Energy released when gaseous ions combine to form a crystal lattice.
- **Factors**: Directly proportional to the product of charges ($q_1 q_2$) and inversely proportional to size (r).

Fajan's Rule

- Describes covalent character in ionic compounds.
- **Conditions for high Covalent Character:** Small cation, large anion, and high charge on ions.

Properties of ionic compound

- **Physical State:** Solid at room temperature.
- **Conductivity:** Non-conductors as solids; good conductors in molten state or aqueous solution.
- **Solubility:** Soluble in polar solvents (like water).

Inter Molecular Force of Attraction

- **Strength Order:** Ion-Ion > Ion-Dipole > Dipole-Dipole > Ion-Induced Dipole > Dipole-Induced Dipole > London (Dispersion) Forces.

Hydration Energy

- Energy released when ions undergo hydration (surrounded by water).
- Smaller ions have higher charge density, leading to higher hydration energy and larger hydrated radii (e.g., $Li^+_{(aq)} > Na^+_{(aq)}$).

Hydrogen Bonding

- Attractive force between Hydrogen attached to a highly electronegative atom (F, O, N) and another electronegative atom.
- **Types:** Inter-molecular (between different molecules, e.g., H_2O) and Intra-molecular (within same molecule, e.g., o-Nitrophenol).